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**A field review of cognitive and psychosocial
factors influencing software development
management.**

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ABSTRACT

Software development projects are prone to systematic failure as a cause of involved parties processing and sharing information in maladaptive ways that impedes progress of the project. Two explanatory ways of information processing are presented. First is the confirmation bias, which describes a natural tendency of individuals to avoid information that is not the information they would like to be seeing. The other is the MUM effect, which is the tendency of individuals to withhold information for various internal and external reasons. An observational case study is presented that describes a software project plagued by the issues that the two factors are responsible for in many cases. Some tips for Information Technology managers are presented with the aim of helping to recognise and battle the consequences of the factors taking effect.

Keywords:

cognitive bias, software project, information transmission, mum effect

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1 Introduction

The cognitive dynamics and social dependencies present within a project environment are not to be taken lightly by any prospective project manager or support worker. Experiments show (Wason, 1960) that people exhibit certain specific patterns of cognition when searching for information, so called cognitive biases. They present an environment where "individuals draw inferences or adopt beliefs where the evidence for doing so in a logically sound manner is either insufficient or absent", as Haselton et al., 2005 note, which can be dangerous in a context such as software development projects. Wason showed in his experiments that people are prone to the mistake of trying to confirm their belief about an issue rather than trying to disprove it which may result in mistakes that cost the project, and the company, resources that could be better spent elsewhere.

Another important dynamic to consider is the social and organisational ties within the company. People notably hesitate to transmit negative information, either when facing subordinates (Manzoni, 2002), when the information is presented such that its transmission would endanger the employment position of the person relaying it (Keil and Robey, 2001) or even in situations where one "finds himself agreeing to promote a position the content or direction of which goes counter to the inferred attitudes of an anticipated audience (and sometimes also counter to his own)" (Rosen and Tesser, 1970) - like for instance promoting a solution that is visibly unprepared for production, unbeknownst to the project financier.

Such mistakes, when committed by project management, are a possible contributing factor to the fact that software projects are not a very stable entity in general, as described by Tan et al., 2003; this analysis of over 20 thousand such projects demonstrated that nearly three quarters of analysed projects failed to deliver the promised product, were over budget or did not deliver at all. It would be reasonable to want to avoid such outcomes, given that they are highly likely to introduce additional costs incurred by the company while effectively siphoning off resources for an unnecessarily long time. This paper aims to observe the aforementioned factors in the field and to propose ways of countering them via the lens of the analysed project while keeping in mind their general applicability.

First, a cognitive bias that describes inefficient information seeking will be introduced, along with practical demonstrations of it. The following chapter will describe in detail the tendency to withhold important information when not doing so would jeopardise the position of the person relaying it. Next, a case study of a software service development project in the field will be presented. The final part of this paper will discuss the implications of the findings and possible defense mechanisms against the behavioural and cognitive information processing threats presented in the first part.

1.1 Cognitive bias

Undeniably, people regularly make mistakes in judgement and thinking that result in unfavourable outcomes. However, there are certain specific and puzzling ways in which the human mind occasionally deviates from what can be rationally considered the most "correct" behaviour. Pohl (2004) presents over 21 such fallacies in his book, noting that some like misconceptions of causality and physics, along with lesser ones, are not even counted. These distortions are natural, as people aren't computers, being able to calculate all possible outcomes and select the best ones. Rather, people tend to use cognitive tools in thinking, judgement and memory storing and recall - so called heuristics. These are "shortcuts or rules of thumb that are prone to breakdown in systematic ways" (Haselton et al., 2005), which is hardly optimal in a software project environment.

Take for instance the well-studied thought experiment 'Linda', a cognitive bias called the conjunction fallacy. In the original experiment done by Tversky and Kahneman (1983), the researchers describe a woman named Linda and list an assortment of attributes about her life, such as the fact that she was intensely interested in issues of social equality when at university or is single and in her thirties. Then it posits the question "What profession is Linda currently engaged in?" to the participants with three choices, either that she is a lawyer, an active feminist or a lawyer and an active feminist at the same time.

Mathematically, the third option is the least probable, because a conjunction of two statements is never more probable than either single statement on its own. What was fascinating however, is that 85% of all participants of the experiment committed the fallacy.

One interesting tendency of people is to seek out information that confirms their reason for searching for it, and being very stringent and hypercritical in response to information that invalidates their belief. This gravitation toward information that only stresses the positive is risky, because for smooth project work flow, all available data needs to be considered. A lot of information is negative in nature in any human endeavour. It is not productive to only regard some information about a project issue and discard the rest, even if unintentionally.

What follows is an outline of this tendency, coined the confirmation bias.

1.1.1 Confirmation bias

"[Towns under Athens' rule] judgment was based more upon blind wishing than upon any sound prevision; for it is a habit of mankind to entrust to careless hope what they long for, and to use sovereign reason to thrust aside what they do not fancy." (Thucydides, The Peloponnesian War, Book 4, chapter 108, section 4)

At its core, the confirmation bias describes the fact that people are often inclined to look for information that supports their view and so they process hypotheses in such a way that the hypothesis is inevitably confirmed. This bias creates an environment where "information is searched for, interpreted, and remembered in such a way that it systematically impedes the possibility that the hypothesis could be rejected" (Pohl, 2004).

In the field of scientific inquiry, this skews and muddies results, but in a business context it leads to revenue loss if the (unknowing) application of confirmation bias leads to project deadline extension or additional funds being funneled into the project. After all, there are no 'do-overs' in live production, as opposed to scientific work which sometimes requires and often warrants multiple experiments of the same kind in order to ensure consistency of the results.

It is, however, possible that confirmation bias is simply a strategy used by people to filter out unlikely outcomes and effectively restricts the set of tests that need to be done. This is in line with the commonly held belief that the reason for heuristics existence in the first place is to overcome evolutionary shortcomings or inadequacies of the brain by limiting the pool of information that needs to be processed, instead of systematically reviewing every single possible case which is not practically feasible.

1.1.2 The Positive Test Strategy

This exploration space restriction is what Klayman and Ha (1987) dubbed the Positive Test Strategy (PTS) - it holds that "you test a hypothesis by examining instances in which the property or event is expected to occur (to see if it does occur), or by examining instances in which it is known to have occurred (to see if the hypothesized conditions prevail)."

PTS theory states that people possess an inclination toward asking questions so that their assumed hypothesis "would be confirmed if the answer was affirmative" (Pohl, 2004). In essence, Pohl is of the opinion that the PTS is an effective search strategy, but it does not necessarily lead to preservation of the original hypothesis. However, people generally only consider their initial hypotheses and do not take into account alternative explanations when applying the PTS.

This leads to yet another challenge in information processing, namely selecting the appropriate hypotheses. Pohl summarises it as such: "Independent of whether we accept or reject a hypothesis, it may be true or not. Thus, with respect to a hypothesis, there are in addition to its correct acceptance and its correct rejection, two kinds of

erroneous decisions: acceptance of a hypothesis even though it is not true (error of commission), and rejection of a hypothesis even though it is true (error of omission)."

The PTS therefore is liable to induce the error of omission and commission at the same time - when the strategy is employed, the employer may leap at the positive explanation and discard the less likely but possibly more stable ones and so very little actual information is gathered via the employer's efforts and their hypothesis is merely confirmed. In search of this result, possible explanations are rejected (error of omission) and the favoured hypothesis is swiftly highlighted as the end point beyond which no further searching is required (possibly error of commission, dependent on the nature of the search).

1.2 Psychosocial argument

Proper technical management is crucial in any project management context and so keeping up to date with technologies or being more involved in the user experience, for instance, is sound advice when trying to minimise risk. But psychosocial factors play an undeniable role too, and some publicised software project failures point, according to Tan et al. (2003), to the fact that misreporting project status was a contributing factor in the demise or at least unnecessary prolonging of the projects. "One major cause of software project failure is the reluctance of people to report bad news about a project and its status. While evidence of failing may be apparent to people involved in a project, this information may not be communicated up the hierarchy or may be substantially distorted in the communication process." (Tan et al., 2003). Mistakes are costly, and Park et al. (2008) show that companies are increasingly turning to outsourcing software solutions and external vendors and so shifting blame becomes easier, which fosters an environment in which bad news reporting may be affected.

1.2.1 The mum effect

We've all experienced the reluctance to say something negative at some point in our lives. Imagine that you have a friend called Tom. Tom has applied for a position at your company and you learn after the interview from your boss that Tom did really well in his interview and they want to offer him a spot working for the company. You just can't wait to tell him! You're bursting with enthusiasm when you meet him, you might even skip the pleasantries

and dive straight for the good news. Now, imagine that your boss told you that he didn't find Tom to be a good fit and the company will not pursue an employment agreement with him. Now you're not so sure whether to tell him or not. Maybe you don't want to bring him down, or maybe you don't want to adopt a negative mood as telling bad news implicitly requires one by society to present them in a somber manner.

This is a rather trivial case, but the implications of it are far-reaching and profound. Not universal, though. The media, for instance, tend to convey bad news with cheerful and brisk readiness, with every major news channel being in effect a dispensary of fear and misinformation. Major crises such as a global pandemic obviously require thorough coverage, but overreporting an issue doesn't do much else other than generate low-grade hysteria in the best case, and contributes to riots and city shutdowns in the worst. At the end of the day however, one may very well argue that this is because "peace and tranquility don't hold any news value" and, after all, news is still a business.

We also don't universally exhibit a bias *against* telling negative information to one another. Tom might not have gotten the job, and we didn't tell him, but we most likely did tell *someone*. We tend to share gossip or urban myths quite freely, which proves that we don't really mind the bad news itself, we just do our best to keep in mind the position of the parties involved in information transmission.

The basis for this phenomenon of being selective about relaying negative information is rooted in the literature of whistle-blowing. Whistle-blowing theory holds that "people choose to disclose (or not disclose) information due to personal, project team, and organizational factors." (Tan et al., 2003). The people who report organisational dysfunctions may not be in a position of sufficient authority to be able to address the problem and so they turn to persons with a higher level of power over the matter in order to lead to an eventual resolution. This is a factor identified by Tan et al. as "organisational climate" and is described as "the extent to which people have an incentive to shirk because their interests diverge from those of the organization." The incentive may be to prevent reprimanding or to appear as more competent to senior management in a quest to reach the higher strata of the organisation's employment ladder. Regardless, withholding relevant information goes counter to the objectives of any project. This tendency is known

as the mum effect, sometimes abbreviated as the M.U.M. effect, which stands for "(keeping) Mum about Undesirable Messages" (Rosen and Tesser, 1970).

One interesting field where the MUM effect is frequently studied is the medical industry, specifically physicians delivering bad news to their patients (with physicians in some cases delivering bad news on a daily basis). These physicians have a vested interest in MUM effect research, namely because they are interested in understanding it so that they may break bad news to their patient such that the patient experiences as little emotional distress as possible as a result of the news being grim. Affected are also the physicians, however, because, as Dibble and Levine (2010) point out, "the stress they experienced from delivering bad news frequently lingered from several hours to 3 or more days". This underlines the importance of relaying bad news correctly, even if there is no apparent immediate consequence in doing so.

The inhibiting effect of keeping mum is not uncommon in software project settings, which the following case study aims to demonstrate.

2 Methodology

Given that the objective of this paper was to observe whether the presented factors would be visible in a real-life scenario, an observational case study approach has been selected. The case being explored is a project dubbed 'Stale Accounts', developed by a large international logistic company, namely it's Information Services division. In the interest of confidentiality, the company will be referred to as 'ZMK'.

In total, 14 interviews with developers, managers, IT support staff and users were conducted. An undetermined number of observations have been made in meetings pertaining, among others, to the Stale Accounts project and during daily operation of the service. Finally, a multitude of email chains, reports and pieces of project documentation have been analyzed.

3 The Stale Accounts project

ZMK is a major logistics company with offices in over 100 countries. It uses the tool ServiceNow to manage their portfolio of services, configuration items, accounts, integrated data systems and more.

In late 2015, ZMK was in need of a piece of software tool that would enable remote management of computer accounts in Active Directory. It has been found that after an employee leaves the company, the account for their computer(s) remains in Active Directory, marked as active. This is in essence a sink for resources, as the number of these accounts that need to be unnecessarily hosted would inevitably rise as a consequence of daily operational matters.

Up until this juncture, ZMK employed a simple ad-hoc process of compiling semi-regular monthly lists of unused computer accounts in Active Directory and then sending them to the respective country and division pairs within the company, to be dealt with operationally at the behest of the division's IT manager. The process worked well enough for small batches of computer accounts, and management decided that it would be very cost-effective to implement an automated solution to replace the manual sending of computer account lists. In fact, the solution concept was so promising to the project financier that it was decided to extend the solution to include user accounts as well, because these accounts are disproportionately more expensive to host than are computer accounts.

Development progressed rapidly and soon a proof of concept was available. But the cost-cutting was too aggressive on management's part and the proof of concept turned out to be very rushed and unfinished. One support worker described the meeting where the proof of concept was being presented as somewhat awkward; the solution looked confusing and quite hard to use, with the principal project manager even informally mentioning outright to the few people present offline that the solution is "bad" (only using a more vulgar synonym).

Nonetheless, the solution was released into production. Only for a short time, however, as it was not tested properly and was still riddled with bugs that made the solution unusable under certain user conditions, in addition to being poorly ergonomic from the user experience perspective.

During this time, ZMK was partnered closely with a ServiceNow certified consulting and development firm, going so far as seating many key employees of the consulting firm inside the ZMK building, right next to internal employees.

The faulty solution was pulled from production so that it could be looked at by the external firm and moved to a usable state. This turned out to be a mistake in this particular situation, as the external developer assigned to fix Stale Accounts still left the service in a faulty state after development, even being visibly unfamiliar with the system and poorly prepared during knowledge transfer sessions. After releasing Stale Accounts to production once again, more bugs, some with no available workarounds, have been introduced. The solution was still in a Hypercare period, which is a time frame of two weeks after release to production is completed during which any potential bugs are fixed on priority and free of any additional service charges. So the solution was pulled from production again and looked at by the external vendor, again.

This curious management decision failed to restore the solution to its original functioning state. Bugs were still very much present, and support staff was beginning to become restless. A poorly designed service is hard enough to debug, let alone support it with end users in mind. Interestingly enough, the solution was still up and running despite its many fundamental flaws, such as marking some accounts for deletion wrong or failing to send crucial notifications to line managers of the owners of these accounts. This led to many instances of data inconsistency, which again only served to provide more grief for the support staff who would eventually be forced to take informal ownership of the service.

These failed restoration attempts were not completely ignored. ZMK would begin with letting go the external developer responsible for the bugs from the company, and eventually proceed to slowly cut ties with the external firm. This partnership is currently dwindling, although some of the external firm's employees are still positioned at ZMK offices and remain in their original posts in some project areas to this day.

Nonetheless, the solution was necessary and useful to management's budget constraints and so it was not abandoned. It remained in a semi-stable, semi-functional state for nearly three years, with no official support team being assigned to assist users with Stale Accounts-related issues. Support staff voiced the opinion that it would be, from their perspective, far more rational to just scrap the solution and start over. This did not come to be. For these three years, the issues mounted and some computer accounts were stuck in the deprovisioning process for months at a time. It didn't seem to matter to management, because most complaints that were received were handled internally with no official structure or evidence of the issue when they came up. Even in the face of some deprovisioning orders failing outright, management decided to customise the solution for each business unit specifically to comply with their varied requirements due to differences in handling computer account storage.

Again, this turned out to be a bad move. The Stale Accounts service was not ready for any modifications, barely being functional on its own. The project manager assigned to the customisation project seemed very keen on pushing the solution through as hard as possible. It was the opinion of his interviewed colleagues that he was attempting to portray a façade of competence and speed in delivery of solutions. A very well paid position was open in the project manager's line manager's team, and presumably the project manager wanted to get this position bad enough that he was willing to withhold the true status of the project to his superiors.

The customisation project was greenlit and funded extensively because this time, the impact of the service was greater than before as it had to comply with various hitherto unknown requirements. As seen in email chains related to this stage of Stale Accounts development, the project manager seemed to be focused in an almost myopic sense on providing the requested modifications for his preferred future division, ZMK EXP, where the attractive open employment position was situated. Some concerns from the Information Services division were put forth, but quickly dismissed by the project manager as being less than important. He was partially blinded by ambitious aspirations of high corporate positions, and the accompanying perks, to the extent that the concerns raised by Information Services support staff, who were at this point (forcibly) well acquainted with the service, were overlooked in the final analysis and so the issues persisted into live operation of the service. No one wanted to get involved with any part

of the service when it was live at this stage. Several support teams were contacted and meetings with the respective team leaders were set up to decide who will receive formal ownership the the service, but they didn't prove fruitful as no team ended up claiming the service for themselves, leaving it in limbo and the hands of disgruntled support staff, who had better things to do than putting bandage-like fixes on a faulty service (that they were not even officially responsible for).

In the span of these three years, approximately 2300 incident tickets were logged by end-users and support staff, all related to Stale Accounts in some fashion.

Eventually, after three years, a support team was found that took ownership and created and distributed the RACI matrix for the solution to everyone involved. It took quite a while for the team to get familiar with the operation of the service. The service was created by a small handful of developers who have in the meantime left the company, taking all practical knowledge out the door with them because the service was seriously lacking in proper documentation. This means that from the outset, no one really knew how the solution worked in detail, not even what server the service was running on. The official support team managed to piece the information together however, and operation of the service became the smoothest it has ever been. This didn't say much though, as the bar for operating smoothness was set very low when the solution was released into production three times in the span of a few weeks during its early stages. The number of tickets opened each month regarding Stale Accounts began to go down as the support team started taking the reigns into their hands and proposed relevant modifications to the service. Such operation continued for over a year and while the issues were most certainly not "fixed", they were handled by the support team well enough to not warrant any immediate action on the part of the solution developers.

At present, the Stale Accounts service, almost five years after its inception and countless costly tickets being logged, is being slowly phased out in favour of a new conceptual ground-up solution intended to completely replace it. The original project manager was promoted to his desired position and the matter was not spoken about again in managerial circles until only very recently.

4 Discussion

Cognitive confirmation bias and the MUM effect serve as adequate explanations for the observed financially irrational behaviour of Stale Accounts project management. The project would have almost certainly been terminated much sooner had management been made fully aware of its status at all stages of its existence, but in the pursuit of cutting costs, the true status was obfuscated and the faulty solution was allowed to continue wasting company resources unfettered.

The MUM effect is thought to have played a role in the project manager's decision-making process. The project manager was seemingly preoccupied with garnering favour with the powers that be at the expense of the quality of the project. Clearly, despite the flaws the project had, everyone who was working directly on the solution was aware of them except for management.

As previously noted, Pohl (2004) conceptualised the confirmation bias as a particular way of searching for and interpreting available data to ascertain a hypothesis that the information seeker wants to confirm. Therefore, one could surmise that the fact that the project was not terminated even while it was essentially a failure could also be due to confirmation bias on the part of management, who focused solely on cutting costs. Other attributes of the project were overshadowed by the fantastic cost reduction it promised to achieve. In the long run, the project clearly went counter to that notion by remaining in a somewhat-broken state for quite some time and proved to be more cost-generating rather than cost-reducing.

We don't have to remain slaves to the unconscious cognitive whims of our and our colleagues' minds. There are ways of exercising vigilance and implementing a sort of checks-and-balances system when dealing with situations in which these factors may endanger the result of our efforts. A few such practical tips are as follows.

4.1 Confirmation bias - ways of protection

Confirmation bias literature is clear on the fact that people are prone to seeking information that confirms their beliefs or expectations. By that token, one obvious strategy would be to force oneself to consider hypotheses that are directly *opposed* to what one holds as implicitly true or what one expects or wishes to happen. Unshakeable belief in the accuracy of one's assessment of the situation at hand is key in perpetuating people's inaccurate information processing. One must detach from lofty expectations that one puts on the efficacy of their proposed solutions to problems, otherwise the damages will snowball until one finally admits that opposing information was overlooked. Nickerson (1998) provides empirical evidence that the approach of training people to think of alternative hypotheses early in the hypothesis-evaluation process shows great promise in blunting the expression of the confirmation bias.

Another prophylactic agent against the nefarious effects of the confirmation bias is simply being mindful of it; mindful of the fact that we are just people, who are frequently wrong regardless of the strength of the belief that we are not. Nickerson puts it like this: "[A] critical step in dealing with any type of bias is recognizing its existence, perhaps simply being aware of the confirmation bias—of its pervasiveness and of the many guises in which it appears—might help one both to be a little cautious about making up one's mind quickly on important issues and to be somewhat more open to opinions that differ from one's own than one might otherwise be."

Another interesting avenue to consider is the fluency of the information being presented. Hernandez and Preston (2013) define fluency as "the relative ease experienced during processing, and can be altered by features such as the visual clarity of text". They show that manipulating the fluency of the information that is being presented affects the depth of thought that is given to the information, that is to say the more disfluent the presentation is, the more cognitive resources are called upon to process the information comprehensively. Therefore, when, for example, making far-reaching managerial decisions regarding a given project, Hernandez and Preston encourage people to make the search and processing of information deliberately more disfluent, or difficult, in order to reach conclusions more in line with what reality holds.

4.2 MUM effect - ways of protection

With respect to the MUM effect, we have the ability to tailor the transmission of information to the situation at hand in order to suppress the negative effects of its transmission. These insights are gained largely from the aforementioned medical industry, which supplies researchers with (quite unfortunately) many cases of bad news being transmitted.

Ptacek and Eberhardt (1996) categorise their recommendations into two distinct sets, physical and social setting (location, structure) and message (what is said, how it is said). If the subordinate is afraid that mentioning bad news to their manager will put their employment position in jeopardy, they can minimise this risk by delivering the news in a private, quiet location, in addition to structuring the delivery such that the time of it is convenient to the manager. Second, it is stressed that one needs to balance the need for complete understanding of the situation by the manager with breaking the news in a way that does not worsen their appraisal of it. Such situations require one to accurately determine how the manager might interpret the information. This is easier said than done. In many ways, "appraising news valence is a subjective enterprise, largely because bad news is defined from the perspective of the receiver. Nonetheless, senders who can maximize their accuracy of inferring the receiver's appraisal should find themselves in a position to better manage the outcomes of the news disclosure." (Dibble and Levine, 2010). In essence, this means that to break bad news effectively (with respect to the subject matter and individual dispositions), the position of the manager needs to be considered carefully.

To counter the reluctance of transmitting negative information itself, Park et al. propose numerous ways of modifying the structure and quality of the superior-subordinate relationship in order to make more probable and facilitate the transmission of negative information. For instance, they showed that time urgency, defined as "the time constraint [that] induces some feeling of stress and creates a need to act within the limited time frame", can be a robust predictor of the willingness of the subordinate to report the bad news. As such, managers can make use of this finding by emphasising project deadlines to their subordinates and stressing their importance.

Another salient point to consider is the manager's implicit beliefs about their subordinates. Morrison and Milliken (2000) argue that there is a predominance of a certain viewpoint about employees present at business schools and consequently in the

general belief systems of individual managers which states that individuals are "self-interested and act in ways to maximize their individual utilities". Managers may come to believe that their subordinates are untrustworthy and therefore they may act to foster (even subconsciously) a company environment that discourages free discourse, because their subjects simply cannot be trusted with such a degree of informational freedom. Morrison and Milliken find that such beliefs are not conscious and so it would serve managers well to voluntarily take a step back and examine the basic workings of the organisational structure they have come to accept. It is important to honestly review one's beliefs to identify any potential areas for improvement and the beliefs managers hold concerning their employees are no different. If a manager feels that he is not getting the full picture from his subordinates or simply wishes to prevent such a thing from happening in the future, it would be fruitful to reassess the view the manager has of the moral fibre of his subjects.

5 Ending remarks

To err is to be human. We should not be ashamed of our natural tendencies, as this impedes their understanding and eventual remedying. We should, however, be cognizant of them at all times.

Erroneous presuppositions are central to the confirmation bias' ability to distort one's view of a given situation, and they work in tandem with a simple lack of awareness of it. One can try to lessen the destructive effects of the confirmation bias by first educating oneself about it and then examining their initial hypotheses before beginning their search for information.

The MUM effect is one based on uncertainty and mistrust. Managers and their subordinates can meet in the middle and work together to foster an environment in which voicing opposing viewpoints is encouraged and then relaying and processing information so that the product being developed takes precedence over individuals' own internal objectives.

The more there is at stake, the more these points hold value for any company. Overlook them at your own peril.

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